Pieces at girth five: the case of bounded girth is not necessary

A second research note on “The maximum number of cycles in a graph of large rank”

Fable 5.2, for KNT — Research note of 12 November 2026

This note answers the second research task of 12 November 2026: can the case of bounded girth (Section 7 of the paper [M]) be removed by a stronger lemma on pieces? The answer is yes. The numbers of the statements of [M] are those of the draft of 42 pages, and [N1] is the first note, `one-scale.pdf`. The manuscript is not changed.

The result in eight sentences

1. *The audit.* Theorem 4.4 of [N1] is correct. Section 2 writes its induction step again, with the two regions and all conditions.
2. *The new theorem.* The lemma on pieces holds at girth five: a connected subcubic multigraph of girth at least five with at most k boundary vertices and at least $C_1 k^7$ vertices has a forest proportion of less than $\frac{3}{5}$ for every even set of boundary vertices (Theorem 4.2).
3. It needs no hypothesis on the cuts of an ambient graph. A cubic graph without a cyclic cut of size at most four has girth at least five, and this is all that the proof uses.
4. *The reason.* A piece with a larger forest proportion is thin. A long thin piece contains a part in which *every* join is a forest. In such a part, a tree that connects the terminals passes one time through a region far from the terminals. If the number of terminals on each side is odd, this region has a full pair of ports. By [M, Lemma 5.8] it is a capped ladder, and a ladder has cycles of length four.
5. *The consequence.* [M, Theorem 8.1] holds for girth five. Then Case 4 in the proof of the main theorem is empty. Section 7 (Theorem 7.1), Lemma 5.5 and Corollary 5.6 are not necessary, and they are used nowhere else (Section 5).
6. *Measured.* Two copies of the paper without Section 7 build. The lean form has 1617 typeset lines and 42 pages (now 1624 and 42). The clean form has 1643 lines and 43 pages. The main theorem has four cases in the place of five.
7. *Tests.* In all families that I computed, the largest forest proportion decreases with the size, and the ladder is the only family with a rigid set of unbounded size (Section 6).
8. *The price.* The constant C_1 is very large, the proof of the lemma on pieces is longer, and the text differs more from the formal proof, which has the case of bounded girth.

> NOTATION | **the status of this note**
> - *Established on paper.* Sections 2 to 5 contain complete proofs. They use [M, Theorem 3.1] through

[M, Lemma 4.1], and [M, Lemmas 5.2 and 5.8]. I checked each step against these statements.

- *Tested by computer.* Section 6 has complete enumerations on small graphs. These tests do not replace a proof, and the constants of the theorem are far outside their range.
- *Not done.* There is no formal proof. No other person or model has read this note.
- *The manuscript.* It is unchanged. The script `apply_girth_five.py` applies either form to a copy.

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1 The setting

A graph may have loops and parallel edges; a multigraph of girth at least five has none. Let N be a connected subcubic multigraph. Its vertices of degree less than three are its *boundary vertices*, and its *deficit* $d(N)$ is the sum of $3 - \deg v$ over all vertices v. A count of edge ends gives

$$|V(N)| = 2\beta(N) - 2 + d(N). \tag{1.1}$$

Let T be a set of boundary vertices of even size. A *T-join* is a set of edges in which exactly the vertices of T have odd degree. There are $2^{\beta(N)}$ of them. The *forest proportion* of T is the proportion of the T-joins that are forests. We call T *rigid* if every T-join is a forest. For two vertices $x \neq y$ of degree at most two, an $\{x, y\}$-join is a path from x to y and some cycles, pairwise disjoint; so the set $\{x, y\}$ is rigid if and only if every $\{x, y\}$-join is a path. This is a *full pair* in [M].

Fine print. Statements that the note uses.

(a) Thin pieces ([M, Lemma 4.1] and the step "A marker" in the proof of [M, Lemma 8.2]). There is an absolute constant $B > 2^{100}$ with the following property. Let N have at most k boundary vertices, let o be one of them, and suppose that some even set of boundary vertices has forest proportion at least $\frac{3}{5}$. Then fewer than $W_k = B(k+1)^2$ vertices have any given distance from o. This step uses no hypothesis on the girth.

(b) Poles ([M, Lemmas 5.2 and 5.8]). A *pole* is a connected multigraph without loops in which every vertex has degree two or three; the vertices of degree two are its *ports*. Let x, y be a full pair of ports of a pole M_1 of rank $b \geq 1$. Replace every other port, with its two edges, by one edge. The result M_1° consists of bridges and of bridgeless multigraphs $N_1, \ldots, N_q$ of ranks $b_i \geq 1$ with $b = \sum_i b_i$. Each N_i has two ports, and they form a full pair. Each N_i is a pair of parallel edges, or a *capped ladder*: a chain of tiles, where a tile is a copy of $K_{2,2}$ (a *square*, of weight 2) or one edge (a *single rung*, of weight 1), two disjoint edges join the right pair of each tile to the left pair of the next tile, and each of the two ports is joined to the two vertices of an end pair. A capped ladder of total weight w has rank $w + 1$.

(c) Separators ([M, Lemma 8.4], [N1, Lemma 1.2]). Pairwise disjoint connected sets that separate two vertices u and p of a connected graph are linearly ordered, $C_1 < C_2 < \cdots < C_n$. For an integer $w \geq 1$ there are pairwise disjoint sets Y_l, one for each $l \geq 0$ with $(l+1)w \leq n - 1$, such that Y_l contains $w - 1$ of the sets C_i,

every edge that leaves Y_l ends in C_{lw+1} or in $C_{(l+1)w+1}$, and Y_l lies in the component of u without $C_{(l+1)w+1}$.

(d) Free regions ([N1, Lemma 4.1]). A region A of N is *T-free* if it contains no vertex of T and every component of $N - A$ contains an even number of vertices of T. If N has n pairwise disjoint T-free cyclic regions with at most d boundary edges each, then the forest proportion of T is at most $\frac{1}{2} + (d+1)2^{d-1}/n$.

(e) Product ([N1, Lemma 4.2], with numbers f_i in the place of $\frac{3}{5}$; the proof is the same). Let $U_1, \dots, U_m \subseteq V(N) \setminus T$ be pairwise disjoint sets of vertices of degree three such that each $N[U_i]$ is connected, and let f_i be a number such that every even set of boundary vertices of $N[U_i]$ has forest proportion at most f_i. Then the forest proportion of T is at most $f_1 \cdots f_m$.

(f) Few short cycles ([N1, Lemma 4.3]). For $k \geq 1$ let s_k be the least integer with $2^{s_k} > (2s_k + 1)W_k$. Let N have at least two and at most k boundary vertices, put $s = s_k$, and let t be the size of a maximal family of pairwise disjoint cycles of length at most $2s$. If N has more than $3 \cdot 2^s(k + 2s\,t)$ vertices, then every even set of boundary vertices has forest proportion less than $\frac{3}{5}$. This lemma has no hypothesis on the girth.

2 The audit of Theorem 4.4 of [N1]

Theorem 2.1 ([N1, Theorem 4.4]). Fix k_0 such that $4s_k < k$ for all $k \geq k_0$, and put $D_* = 2s_{k_0} + 1$. There is an absolute constant C_1 with the following property. Let $k \geq 1$, and let N be a connected subcubic multigraph of girth at least D_* with at most k boundary vertices and at least $C_1 k^7$ vertices. Then every even set of boundary vertices has forest proportion less than $\frac{3}{5}$.

The result of the audit. The proof is correct. One passage was not precise: the proof took "one of the others" and then used "two sets Y'". The step is written in full below, and [N1] has the new wording. I found no other defect. The proof uses the girth in two places only. These are the base, and the number of vertices of a short cycle in (6) below.

The numbers. For $k < k_0$ put $F(k) = 3 \cdot 2^{s_k} k + 1$. For $k \geq k_0$ put $s = s_k$, $k' = 4s < k$ and

$$w = \lceil k' F(k')/D_* \rceil + 1, \qquad n_1 = 5(2s+1)\,2^{2s}, \qquad n_2 = k^2(k+2)\,w,$$
$$F(k) = 3 \cdot 2^s \big(k + 2s\,(k + n_1 + n_2 + 2)\big) + 1.$$

The claim $P(k)$ is: a multigraph N as in the theorem with at least $F(k)$ vertices has forest proportion less than $\frac{3}{5}$ for every even set T of boundary vertices. The recursion is sound, because $F(k)$ uses $F(k')$ with $k' < k$ only.

The base, $k < k_0$. Here $2s_k \leq 2s_{k_0} < D_*$, so N has no cycle of length at most $2s_k$, and (f) with $t = 0$ gives $P(k)$. If N has at most one boundary vertex, then T is empty and N contains a cycle, so the forest proportion is at most $\frac{1}{2}$.

The step, $k \geq k_0$. Let $\mathcal{D}$ be a maximal family of pairwise disjoint cycles of length at most $2s$, and let t be their number.

(1) If $t < k + n_1 + n_2 + 2$, then $F(k) > 3 \cdot 2^s(k + 2s\,t)$, and (f) gives the claim.

(2) Let $t \geq k + n_1 + n_2 + 2$. At most k cycles of $\mathcal{D}$ contain a boundary vertex. Each other cycle Q has at most $2s$ boundary edges, and $V(Q)$ contains no vertex of T. So more than n_1 of

them are T-free, or more than n_2 of them are not.

(3) *More than n_1 free cycles.* By (d) with $d = 2s$, the forest proportion is less than $\frac{1}{2} + (2s + 1)2^{2s-1}/n_1 = \frac{3}{5}$.

(4) *More than n_2 cycles that are not free.* For such a cycle Q, some component of $N - V(Q)$ contains an odd number of vertices of T. As $|T|$ is even and T avoids Q, there are $a, b \in T$ in different components of $N - V(Q)$. There are fewer than k^2 pairs, so more than $(k+2)w$ of these cycles separate the same two vertices a and b. Their vertex sets are pairwise disjoint and connected, and they avoid a and b. By (c), with $u = a$ and $p = b$, they are linearly ordered, and there are $k + 2$ pairwise disjoint sets $Y_0, \dots, Y_{k+1}$.

(5) *Two sets.* At most k of the sets Y_l contain a boundary vertex of N. So there are two indices $l_1 \neq l_2$ such that $Y_1 := Y_{l_1}$ and $Y_2 := Y_{l_2}$ contain none. These two sets are disjoint. Their vertices have degree three in N, and they are not in T.

(6) *Size.* Each Y_i contains $w - 1$ of the cycles, and a cycle has at least D_* vertices. So $|Y_i| \geq (w - 1)D_* \geq k'F(k')$.

(7) *Boundary.* An edge that leaves Y_i ends in one of two of the cycles, and each of them has at most $2s$ boundary edges. So at most $4s = k'$ edges leave Y_i. The set Y_i does not contain b, and N is connected. So every component of $N[Y_i]$ has an edge that leaves Y_i, and there are at most k' components.

(8) *The two regions.* Let Y_i' be the vertex set of a largest component of $N[Y_i]$. Then $N[Y_i']$ is connected, $|Y_i'| \geq F(k')$, and $Y_1' \cap Y_2' = \emptyset$. A vertex of Y_i' has degree less than three in $N[Y_i']$ only if it is an end of an edge that leaves Y_i. So $N[Y_i']$ has at most k' boundary vertices. Its girth is at least D_*.

(9) *Induction.* By $P(k')$, every even set of boundary vertices of $N[Y_i']$ has forest proportion less than $\frac{3}{5}$, for $i = 1, 2$.

(10) *Product.* The sets $U_1 = Y_1'$ and $U_2 = Y_2'$ satisfy the hypotheses of (e): they are disjoint, they avoid T, their vertices have degree three in N, and they induce connected multigraphs. So the forest proportion of T is at most $\frac{9}{25} < \frac{3}{5}$.

The size of F. We have $2^s = O(k^2 \log k)$ and $s = O(\log k)$, so $F(k) = O\big(k^6 \log^5 k + k^5 \log^3 k \cdot F(4s_k)\big)$ for $k \geq k_0$. Let $k_1 \geq k_0$ be such that this bound, with $F(4s_k) \leq C(4s_k)^7$, is at most Ck^7 for every $C \geq 1$ and $k \geq k_1$. Then $C_1 = \max_{k<k_1} F(k)$ serves. ∎

3 Rigid sets at girth five

The ladder with n rungs has girth four, four boundary vertices and a rigid set: the two ends of one rail. So a multigraph of girth four with a rigid set can be large. At girth five it cannot.

Lemma 3.1 (Full pairs at girth five). Let M be a connected subcubic multigraph of girth at least five. Let $x \neq y$ be vertices of degree at most two such that every $\{x, y\}$-join of M is a path. Then $|V(M)| < 15\, d(M)$.

Proof. Put $d = d(M)$ and $b = \beta(M)$. If $b = 0$, then M is a tree and $|V(M)| = d - 2$. Let $b \geq 1$.

A pole. Delete a vertex of degree one other than x and y, as long as there is one. Such a vertex is on no join, so the joins and the rank do not change. If then x has degree one, every join contains the edge at x. Delete x and take its neighbour x' in the place of x. The joins for x' and y are the old joins without that edge, so they are paths, and the rank is the same. The vertex x' is not y: otherwise the old joins are the edge xy together with an even subgraph, all even subgraphs are empty, and $b = 0$. Do the same for y, and repeat. The result M_1 is a pole in which the two vertices are ports and form a full pair, and $\beta(M_1) = b$.

The ports. Let u be a port of M_1. If u has degree three in M, one of its edges leads to a deleted tree. The vertices of this tree have degrees in M that add up to twice the number of its edges plus one. So the tree has a deficit of at least two. Different ports have different trees. Hence M_1 has at most d ports.

The ladders. Let M_1° and $N_1, \dots, N_q$ be as in (b). An edge of M_1° is a path of M_1 whose inner vertices are ports. Call it *plain* if this path has no inner vertex. At most $d - 2$ edges are not plain. A cycle of M_1° with at most four edges, all plain, would be a cycle of M. So every cycle of M_1° with at most four edges has an edge that is not plain. Let e_i be the number of edges of N_i that are not plain. If N_i is a pair of parallel edges, then $e_i \geq 1 = b_i$. Let N_i be a capped ladder with m tiles. Each square is a cycle of length four. Two consecutive single rungs and the two edges between them form a cycle of length four. These cycles are pairwise edge-disjoint, and there are at least $(m-1)/2$ of them, because a run of j consecutive single rungs gives $\lfloor j/2 \rfloor$ cycles and there are at most one more run than squares. A cap and the first tile contain a cycle of length three or four. So $e_i \geq \max\{1, (m-1)/2\} \geq m/3$. The rank of N_i is at most $2m + 1 \leq 6e_i + 1 \leq 7e_i$. Hence

$$b = \sum_i b_i \leq 7 \sum_i e_i \leq 7(d-2),$$

and (1.1) gives $|V(M)| = 2b - 2 + d < 15d$. ∎

Lemma 3.2 (Rigid sets). Let M be a connected subcubic multigraph of girth at least five with deficit d. Let $\lambda : V(M) \to \mathbb{Z}$ be a map with $|\lambda(u) - \lambda(v)| \leq 1$ for every edge uv, such that fewer than W vertices have any given value. If some even set U of boundary vertices is rigid, then $|V(M)| \leq 500\, dW(W + d)$.

Proof. If $U = \emptyset$, every even subgraph is a forest, so M has no cycle and $|V(M)| = d - 2$. Let $U \neq \emptyset$.

A tree. Let S be a connected subgraph of M that contains U and has as few edges as possible. It is a tree, and its vertices of degree one are in U. Call a vertex of S *special* if it is in U or has degree three in S. There are at most $2|U| - 2$ special vertices. A *segment* is a path of S between two special vertices without a special inner vertex. The tree S is the union of at most $2|U| - 3$ segments.

A region on a long segment. Put $\rho = 30W + 8d$ and $\ell_0 = (2\rho + 1)W + 2$. Suppose that some segment σ has at least $2\ell_0$ edges, and let z_0 be a vertex of σ with at least ℓ_0 edges of σ on each side. Let B be the vertex set of the component of z_0 in the subgraph of M induced by the vertices v with $|\lambda(v) - \lambda(z_0)| \le \rho$. Then $M[B]$ is connected, and B has fewer than $(2\rho + 1)W = \ell_0 - 2$ vertices. The values of λ change by at most one along an edge. So B contains the $2\rho + 1$ vertices of σ nearest to z_0.

The tree meets B in one part of σ. Let v be a vertex of S in B, and let π be a path from v to z_0 in $M[B]$. It has fewer than $\ell_0 - 2$ edges. Suppose that v is not an inner vertex of σ. Delete from S the part of σ between z_0 and the end of σ on the side of v, and add π. The result is connected, it contains U, and it has fewer edges, because the deleted part has at least ℓ_0 edges. This is not possible. In the same way, if v is an inner vertex of σ, then σ has at most as many edges between v and z_0 as π has. So the vertices of S in B are inner vertices of σ, and B contains no vertex of U.

Let z' and z'' be the ends of σ, and let S^- and S^+ be the components of $S - B$ that contain them. They are different, because the path of S from z' to z'' is σ, which contains z_0. Every special vertex is joined to z' or to z'' by a path of S without inner vertices of σ. So every vertex of U is in S^- or in S^+. Let x^- be the first vertex of σ in B, seen from z', and let x^+ be the first one seen from z''. Then $x^- \ne x^+$, the vertex x^- has a neighbour in S^-, and x^+ has a neighbour in S^+. Let K^- and K^+ be the components of $M - B$ that contain S^- and S^+, and put $U^- = U \cap V(S^-)$.

The set B is small. We claim that $|B| < 15\, d(M[B])$.

First case: $K^- = K^+$, or $|U^-|$ is even. Then $M - B$ contains a U-join: in the connected set $K^- = K^+$, or in K^- for U^- and in K^+ for $U \setminus U^-$. Its union with a cycle of $M[B]$ would be a U-join that is not a forest. So $M[B]$ is a tree, and $|B| = d(M[B]) - 2$.

Second case: $K^- \ne K^+$ and $|U^-|$ is odd. Let J be an $\{x^-, x^+\}$-join of $M[B]$. Let v^- be a neighbour of x^- in K^- and v^+ a neighbour of x^+ in K^+. The set U^- with v^- added or removed is even, so K^- has a join for it; in the same way K^+ has a join for $U \setminus U^-$ with v^+ added or removed. These two joins, the edges x^-v^- and x^+v^+, and J form a U-join of M. It is a forest, so J is a forest, and hence a path. The vertices x^- and x^+ have degree at most two in $M[B]$. So the claim is Lemma 3.1 for $M[B]$.

An edge of M that leaves B starts in a vertex v with $|\lambda(v) - \lambda(z_0)| = \rho$, because B is a component. There are fewer than $2W$ such vertices, and each has at most two such edges. So $d(M[B]) < 4W + d$, and the claim gives $|B| < 60W + 15d < 2\rho + 1$. This contradicts $|B| \ge 2\rho + 1$. So every segment has fewer than $2\ell_0$ edges.

The count. So S has fewer than $4|U|\,\ell_0 \le 4d\,\ell_0$ vertices. The set $V(S)$ is connected and contains U. So $M - V(S)$ is a forest, as in the first case. Let it have f vertices. At most $|V(S)| + 2$ edges join it to S, and a count of the edge ends in the forest gives $f \le |V(S)| + d$. Hence $|V(M)| \le 8d\,\ell_0 + d = 8d\big((60W + 16d + 1)W + 2\big) + d \le 500\, dW(W + d)$. ∎

Fine print. The proof has three ideas. A rigid set stays rigid in a part of the multigraph, for every way in which the joins can cross the boundary of the part. A tree that connects the terminals with the fewest edges passes through a region around a far point of the tree one time, so this region has exactly two sides with terminals. If both sides are odd, one strand passes, and the region has a full pair; if they are even, a join can avoid the region, and the region has no cycle.

4 Pieces at girth five

Theorem 4.1 (Pieces with few boundary vertices). For every integer $k \geq 1$ put $W = B(k+1)^2$, $h = 10^4\, W^3$ and $F_0(k) = Wh\,(k + 2^{Wh} + 2)$. Let N be a connected subcubic multigraph of girth at least five with at most k boundary vertices and at least $F_0(k)$ vertices. Then every set of boundary vertices of even size has forest proportion less than $\frac{3}{5}$.

Proof. Suppose that some even set T of boundary vertices has forest proportion at least $\frac{3}{5}$. If T is empty, then N has no cycle. A tree with at most k boundary vertices has fewer than $2k$ vertices. So let $T \neq \emptyset$. Fix a boundary vertex o, and call the set of the vertices at distance j from o the layer j.

Thin. By (a), each layer has fewer than W vertices. So N has at least $h\,(k + 2^{Wh} + 2)$ layers.

Stretches. Divide the layers $1, 2, \ldots$ into groups of h consecutive layers. There are at least $k + 2^{Wh}$ groups all of whose layers are not empty, and at least 2^{Wh} of them contain no boundary vertex. Let Γ be the union of the layers of such a group, and let v be a vertex in its last layer. Let M be the component of $N[\Gamma]$ that contains v. A shortest path from o to v has one vertex in each layer, and its vertices in Γ are in M. So $|V(M)| \geq h$. The vertices of M have degree three in N. An edge of N that leaves M ends outside Γ, so it starts in the first or in the last layer of the group; and each vertex of M has a neighbour in M. So $d(M) < 4W$. By (1.1) the rank of M is less than Wh.

Not rigid. The distance from o in N is a map λ on $V(M)$ as in Lemma 3.2. If some even set of boundary vertices of M were rigid, the lemma would give $|V(M)| < 500 \cdot 4W \cdot W \cdot 5W = h$. So for every even set of boundary vertices of M, at least one of the fewer than 2^{Wh} joins is not a forest, and the forest proportion is at most $1 - 2^{-Wh}$.

Product. The sets $V(M)$ for the different groups are pairwise disjoint, they avoid T, and their vertices have degree three in N. By (e), the forest proportion of T is at most $(1-2^{-Wh})^{2^{Wh}} < \frac{1}{2}$. This is a contradiction. ∎

Theorem 4.2 (Pieces at girth five). There is an absolute constant C_1 with the following property. Let $k \geq 1$, and let N be a connected subcubic multigraph of girth at least five with at most k boundary vertices and at least $C_1 k^7$ vertices. Then every set of boundary vertices of even size has forest proportion less than $\frac{3}{5}$.

Proof. Let k_0 be as in Section 2. For $k < k_0$ put $F(k) = F_0(k)$. For $k \geq k_0$ define $F(k)$ as in Section 2, with 5 in the place of D_* in the definition of w. The claim $P(k)$ is the statement of the theorem with $F(k)$ in the place of $C_1 k^7$. For $k < k_0$ it is Theorem 4.1. For $k \geq k_0$ the step of Section 2 holds word for word: (1) uses (f), which has no hypothesis on the girth; (6) holds because a cycle has at least five vertices; and in (8) and (9) the girth of $N[Y_i']$ is at least five. The bound for $F(k)$ is the same, with $C_1 = \max_{k<k_1} F(k)$. ∎

Corollary 4.3 (Two regions)**.** Let G be a cubic multigraph, and let P be a set of vertices such that $G - P$ has girth at least five. Suppose that $V(G) \setminus P$ contains two disjoint sets Y_1, Y_2 and that, for some $k_1, k_2 \geq 1$, each $G[Y_i]$ is connected, $|\delta Y_i| \leq k_i$ and $|Y_i| \geq C_1 k_i^7$. Then $q(G, P) \leq \frac{9}{25}$.

Proof. Theorem 4.2 and (e) with $T = \emptyset$; the proof of (e) does not change when the forest is in $G - P$. ∎

The size of the constant. For $k < k_0$ the number $F_0(k)$ is about $2^{10^4 B^4 (k+1)^8}$, and $B > 2^{100}$. So C_1 is a very large absolute constant. The proof of [M, Theorem 8.1] needs a bound $F(k) \leq 2^{\varepsilon k}$ for the pieces with $k \approx \log r$ boundary vertices; Theorem 4.1 alone does not give this, and the step of Section 2 is necessary for it.

5 The last case of [M] without bounded girth

Theorem 5.1 ([M, Theorem 8.1] for girth five)**.** There is an integer r_5 with the following property. Let G be a connected cubic multigraph of rank $r \geq r_5$. Let P be a nonempty set of at most $3 \log r$ vertices such that $G[P]$ is connected. If $G - P$ has girth at least five, then $q(G, P) < \frac{2}{5}$.

Proof. The proof is the second form of [N1] ([N1, Section 4.3]), with Theorem 4.2 in the place of [N1, Theorem 4.4] and with 5 in the place of D_*. I checked the four applications.

Blocks and the chain. Let A, u, p and $C_1, \dots, C_T$ be as in [N1, Proposition 2.1], with $T > (\Lambda + 2)\big(\lceil 2^8 C_1 \Lambda^8 / m \rceil + 1\big)$. The sets Y_l of (c) avoid P. At most Λ of them contain a vertex with a neighbour in A, and two others remain. For such a set Y, at most 2Λ edges leave Y, and $|Y| \geq 2\Lambda \cdot C_1 (2\Lambda)^7$. So a component of $G[Y]$ has at least $C_1 (2\Lambda)^7$ vertices and at most 2Λ boundary vertices, and it is a subgraph of $G - P$, so its girth is at least five. Corollary 4.3 gives $q(G, P) \leq \frac{9}{25}$.

Heavy families. The proof of [N1, Proposition 2.2] uses the chain and [M, Lemma 8.6] only. The bound is $\mu \leq \frac{3}{2} K + 3\Lambda\kappa\,(\Lambda + 3)\big(\lceil 2^8 C_1 \Lambda^8 / m \rceil + 1\big)$.

Packing. Let $d \geq 5$, and let $Q_1, \dots, Q_n$ be pairwise disjoint cycles of length at most d in $G - P$. Each has at least five vertices and at most d boundary edges. With $\Lambda = d$, $m = 5$, $\kappa = 1$ and

> $K = d + 1$,
>
> $$\mu \le \tfrac{3}{2}(d+1) + 3d\,(d+3)\big(\lceil 2^8 C_1 d^8/5\rceil + 1\big) \le 2^9 C_1\, d^{10}, \qquad n \le 2^{d+1}\mu \le 2^{10} C_1\, 2^d d^{10}.$$
>
> *The final step.* With $D = \lceil x/16\rceil$ and $L = \lfloor Ax\rfloor$, the deleted set has $|S| \le 2^{10} C_1\, 2^D D^{11} \le r/x^2$ vertices. [M, Proposition 3.5] has no hypothesis on the girth of $G - P$. The family of the cycles of $\mathcal{C}$ gives $\mu \le \frac{3}{2}(1+\eta) + 3L\eta\,(L+3)\big(\lceil 2^{12} C_1 L^8/x\rceil + 1\big) = \frac{3}{2} + o(1)$, because $\eta = r^{-1/60}$ and C_1 is a constant. This contradicts $\mu > 30$. ■

The proof of the main theorem with four cases. In [M, Section 9], after the cases of a loop, of a cyclic cut of size at most three, and of a cyclic cut of size four, the multigraph G has no loop and no cyclic cut of size at most four, and $r \ge 9$. Let Z be a cycle with at most four edges. At most four edges leave $V(Z)$, and at most eight edges have an end in $V(Z)$. So $G - V(Z)$ has rank at least $r - 8 > 0$, and $\delta V(Z)$ is a cyclic cut of size at most four. Hence G has girth at least five. Choose eight vertices P such that $G[P]$ is connected. Then $G - P$ has girth at least five, $q(G, P) < \frac{2}{5}$ by Theorem 5.1, and the count of [M, Section 9] gives $c(G) < (\frac{2}{5} + \frac{1}{32})2^r < 2^{r-1}$. So

$$G \text{ has no cyclic cut of size at most four}, \quad r \text{ large} \quad \Longrightarrow \quad c(G) < 2^{r-1},$$

and Case 4 of [M], with a cycle shorter than D_*, is not a separate case.

What is not necessary any more. I searched the source of [M] for each label.

Statement of [M]	Where it is used
Theorem 7.1 (bounded girth), all of Section 7; 52 typeset lines	Case 4 of Section 9 only.
Corollary 5.6 (an induced pole with few ports has not many paths)	The proof of Theorem 7.1 only.
Lemma 5.5 (four thin layers); with Corollary 5.6, 38 typeset lines	The proof of Corollary 5.6 only.
The number D_*, the fifth question of Figure 2, the sixth of the eight sentences	The statement of the cases.

These statements stay: Proposition 5.3 and Theorem 5.4 (Section 6 uses them), Corollary 3.7 (Section 6 uses it), and Lemmas 5.2 and 5.8, which Section 8 now uses as well.

What is new in the text. Lemma 3.1, Lemma 3.2 and Theorem 4.1, with (e) for joins. There are two forms of Section 8.

- *The clean form.* One lemma on pieces, Theorem 4.2. The chain, the heavy families and the packing bound are as in the second form of [N1]; there is no packing number in the statements and no induction in the packing bound.
- *The lean form.* [M, Lemma 8.2] stays, for pieces with at least $2\Lambda_0$ boundary vertices and no short cycle. Theorem 4.1 treats the pieces with fewer boundary vertices. The rest is the first form of [N1]: the chain counts the blocks with a short cycle, and the packing bound uses induction on the length. A block needs $\Phi(\Lambda) = C_1\Lambda^4$ vertices for $\Lambda \ge \Lambda_0$ and a constant

number C_2 of vertices for $\Lambda < \Lambda_0$. The lean form does not need (d), (f) and the step of Section 2.

The size, measured. I applied each form to a copy of the paper and built it. The count is in lines of the text of the PDF file.

	Now	[N1], first	[N1], second	Girth five, lean	Girth five, clean
Section 5 (poles)	339	339	339	301	301
Section 7 (bounded girth)	52	52	52	–	–
Section 8 (the last case)	271	259	289	358	384
The paper, typeset lines	1624	1615	1644	1617	1643
The paper, pages	42	42	42	42	43
Cases of the main theorem	5	5	5	4	4
Statements in Sections 5, 7, 8	20	20	21	20	21

So the removal of the case of bounded girth costs no length in the lean form: the paper has 7 lines less than now, the same number of statements, one section less and one case less. The last section is longer by 87 lines, and Sections 5 and 7 lose 90 lines. The mathematical gain is in the structure: one argument treats every graph without a small cyclic cut. The total amount of proof is about the same.

6 Is the statement true? Tests

The task asks for a search for counterexamples. I found none. All numbers below are complete enumerations of the joins of small graphs by the program `fpmax.c`.

The ladder. At girth four the ladder is a counterexample of every size: the two ends of one rail form a rigid set (computed up to 16 rungs). Its part in the proof is visible: a ladder is the capped ladder of [M, Lemma 5.8], and Lemma 3.1 fails for it because its cycles of length four have plain edges only.

Families of girth at least five. For each graph the table gives the largest forest proportion over all nonempty even sets T of boundary vertices.

Family	k	Largest forest proportion, by the number n of vertices
Segments of generalised Petersen graphs with step 2 (girth 5)	6	1 ($n = 12$), 0.78 (24), 0.64 (32), 0.59 (36), 0.48 (44)
The same with step 3 (girth 6)	8	1 (16), 0.94 (24), 0.84 (32), 0.73 (40)
Rings of length 5 with two and three edges between them in turn (girth 5; cuts of size four)	4, 5	1 (10), 0.88 (20), 0.81 (30), 0.75 (40)
Tubes: rings of length 6, 8, 10 with 3, 4, 5 edges between consecutive rings (girth 6)	6, 8, 10	length 6: 1 (18), 0.84 (36); length 8: 0.99 (24), 0.76 (40); length 10: 1 (30), 0.91 (40)
Rings over the Heawood graph, 7 edges between them; three rings; sets T of size at most 4	14	0.62 (42)
Petersen, Heawood, Pappus, dodecahedron, Desargues, McGee, Tutte–Coxeter without one vertex	3	from 0.88 (9) to 0.55 (19) and 0.62 (29)
Random cubic graphs of girth at least 5 without two or three vertices of a path	4, 5	0.97 to 1 ($n \le 12$), 0.74 to 0.89 (23, 24), 0.68 to 0.81 (31, 32)

In each family the proportion decreases when the graph grows. No graph in these runs with more than 30 vertices has a rigid set. The largest rigid set that I saw at girth at least five is the set of all ten boundary vertices in a tube of three rings of length ten, with 30 vertices and deficit 10.

What the numbers say about the threshold. A piece with 4 to 10 boundary vertices and about 40 vertices can have a forest proportion above $\frac{3}{5}$. So the true threshold is not very small. The theorem gives the exponent 7 with a very large constant. The tests are far from the range of the theorem. They can show an error of the statement at small size, and they show none. They cannot confirm the exponent.

A search for large rigid sets. Rigid sets stay rigid when an edge or a vertex of degree one is deleted. So every multigraph with a rigid set arises from one edge when we add vertices of degree one and edges, through multigraphs with rigid sets. The script `rigid_search.py` takes random steps of this kind and keeps at most k boundary vertices at every step. At girth four it reaches the cap of the rank, 14, for $k = 4, 5, 6$; it finds ladders. At girth five it finds rank 0, 5 and 8 for $k = 4, 5, 6$, with two seeds. This search space is small, so the result is weak evidence.

The reduction. The script `rigid_reduction_check.py` tests the two cases in the proof of Lemma 3.2 for arbitrary connected sets B without a terminal: on 290 rigid sets of small graphs and 7272 sets B, a set with only even sides induces a forest, and a set with exactly two odd sides has full pairs between them. It also tests Lemma 3.1 on the 11420 full pairs at girth five that occur; the largest ratio $|V|/d$ is 1.4.

7 The questions of the task

1. The audit of Theorem 4.4. Section 2. The proof is valid, and the theorem is a tool: Theorem 4.2 is its form for girth five.

2. The theorem for pieces of a graph without a small cyclic cut. It is true, with the exponent $a = 7$ and an absolute constant (Theorem 4.2). The piece N need not inherit any connectivity: the proof uses only that N has girth at least five. I found no counterexample. The two kinds of counterexample are different. A counterexample to the bound $\frac{3}{5}$ is a family of pieces with a bounded number of boundary vertices, of unbounded size, with a forest proportion of at least $\frac{3}{5}$; Theorem 4.1 excludes it at girth five, and the ladder is such a family at girth four. A counterexample to the polynomial threshold is a family with a forest proportion of at least $\frac{3}{5}$ whose size grows faster than every power of k; the step of Section 2 excludes it. The pieces with about 40 vertices in Section 6 are neither: they are below the threshold.

3. Can the induction start at girth five? Yes. The base is Theorem 4.1. A large piece with a fixed number of boundary vertices and a high forest proportion is thin, by (a). A long thin piece contains a stretch with a rigid set, by the product. A rigid set gives a region with a full pair (Lemma 3.2), and this region is a capped ladder up to its ports (Lemma 3.1). So the only obstruction is the ladder, and it contradicts girth five. No small cyclic cut is necessary for the contradiction. The proof does not use [M, Corollary 5.6]. It uses the exact statement [M, Lemma 5.8] on full pairs, and the step from all even boundary sets to one pair of ports is Lemma 3.2.

4. Does this remove Section 7? Yes (Section 5). Theorem 7.1, Lemma 5.5 and Corollary 5.6 are not necessary, and Corollary 5.6 is used nowhere else.

5. If girth five is false. It is not false. For girth four the ladder is the obstruction, and a graph without a cyclic cut of size at most four has no cycle of length four.

8 Limits

- *The constant.* C_1 is the largest value of $F(k)$ over a range of small k, and there $F(k)$ is about $2^{10^4 W^4}$ with $W = B(k+1)^2$. This has no effect on the main theorem, which holds for large rank only. The cause is the product in the proof of Theorem 4.1: it takes 2^{Wh} stretches. I found no polynomial argument for few boundary vertices.
- *The amount of proof.* The removal of Section 7 does not make the paper shorter by much, because Lemma 3.1, Lemma 3.2 and Theorem 4.1 replace it. The proof of Theorem 7.1 with its two dependencies has 90 typeset lines, and the new statements have 87 lines in the lean form.
- *The formal proof.* The Lean proof has the case of bounded girth and pieces of large girth. With a new Section 8 the text and the formal proof differ in the whole last case. The main statement is the same.
- *Dependencies.* Section 8 then depends on [M, Lemmas 5.2 and 5.8]. Before, it depended on Sections 2 to 4 only.
- *No second reader.* The proofs of Lemma 3.1, Lemma 3.2 and Theorem 4.1 are new. They are

short enough to check by hand, and I ask for this check before the manuscript changes.

9 Files and checks

All files are in `paper/minimal/`.

File	Content and result
`research/apply_girth_five.py`	Applies the girth-five form to a copy of the paper: `clean` or `lean`. It writes Section 8, removes Section 7, Lemma 5.5 and Corollary 5.6, and changes Sections 1 and 9. Both copies build with no undefined reference and no overfull line.
`research/trial-girth-five-lean.pdf` `research/trial-girth-five-clean.pdf`	The two copies. The last case starts on page 31.
`checks/girth5/fpmax.c`	Forest proportions of all even boundary sets of a small graph, by enumeration of the joins.
`checks/girth5/families.py`	The table of Section 6. Exit code 0.
`checks/girth5/rigid_search.py`	The search for rigid sets. Seeds 1 and 2, 6000 steps.
`checks/girth5/rigid_reduction_check.py`	The two cases of Lemma 3.2 and the bound of Lemma 3.1 on small graphs. Exit code 0.

The choice. The lean form is the smaller change and the shorter text, and it keeps [M, Lemma 8.2]. The clean form has one lemma on pieces and simpler statements after it. Each of them removes the case of bounded girth. My recommendation is the lean form, after a second reader has checked Section 3 and Theorem 4.1.

References

[M] KNT, Astra and Fable 5.2, *The maximum number of cycles in a graph of large rank*, draft of 12 November 2026, 42 pages, file `paper/minimal/exact-maximum/exact-maximum.pdf`.

[N1] Fable 5.2, *The large-girth theorem with one family of short cycles*, research note of 12 November 2026, file `paper/minimal/research/one-scale.pdf`.